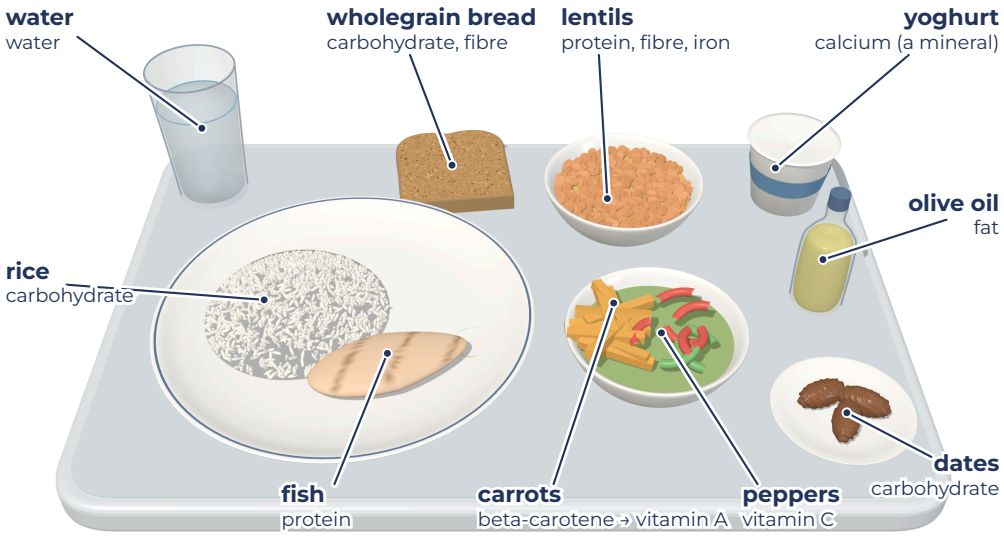
# Balanced diet: reference

Balance means suitable amounts for your needs over time, not equal amounts. Foods supply several components, so these are useful examples.




| Component | Useful function | Useful source |
|---|---|---|
| **Carbohydrate** | Energy for respiration | Rice, bread, potatoes |
| **Protein** | Growth and repair materials | Beans, lentils, eggs, fish |
| **Fat / lipid** | Energy store; insulation; vitamin absorption | Oil, nuts, seeds |
| **Vitamins** | Specific processes in small amounts | C: peppers; A precursor: carrots |
| **Minerals** | Specific processes, e.g. iron for haemoglobin | Iron: lentils; calcium: dairy or fortified alternatives |
| **Fibre** | Helps food move through the gut | Wholegrains, fruit, vegetables |
| **Water** | Medium for many reactions; transport; cooling | Drinking water; water in foods |

| Too little | Body process and possible effect | Useful source |
|---|---|---|
| **Iron** | Haemoglobin helps transport oxygen. Deficiency can cause iron-deficiency anaemia. | Lentils, beans, meat |
| **Vitamin A** | Needed for normal vision. Low intake can impair night vision; severe, prolonged deficiency can cause blindness. | Eggs; carrots (beta-carotene) |
| **Vitamin C** | Healthy connective tissue and wound healing. Severe deficiency can cause scurvy. | Peppers, citrus fruit |
| **Vitamin D** | Helps control calcium and phosphate for healthy bones. Deficiency can cause rickets in children. | Oily fish, egg yolk, fortified foods |

**Starvation:** not enough energy over a long time. **Malnutrition:** too little, too much or the wrong balance of energy or nutrients. Starvation is one form of malnutrition.

Not all anaemia is caused by low iron, and rickets can also involve low calcium. The cases are fictional evidence exercises, not medical diagnoses, and this sheet gives no personal diet advice.

cut here

# Practise, keep, improve

Name _____ Class _____

Cover the reference. Keep first answers, then correct only wrong links in another pen.

## PART A Component links

| Component | Useful function | Useful source |
|---|---|---|
| **Carbohydrate** | | |
| **Protein** | Growth and repair materials | Beans, lentils, eggs, fish |
| **Fat / lipid** | | |
| **Vitamins** | | |
| **Minerals** | | |
| **Fibre** | | |
| **Water** | | |

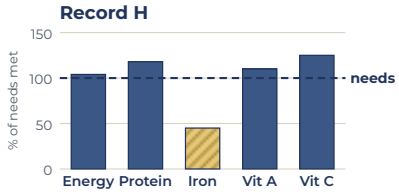
## PART B Four possible effects

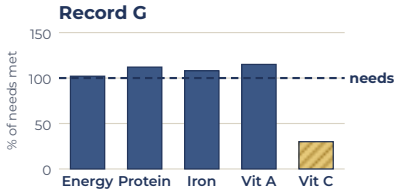
Low iron → _____ Low vitamin A → _____

Low vitamin C → _____ Low vitamin D, child → _____

## PART C Read the records

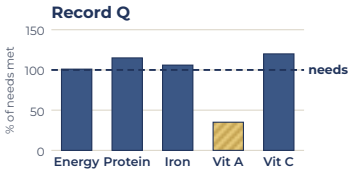
Fictional diet records, averaged over months. Lentils provide iron; peppers provide vitamin C.




| Record | Nutrient gap | Possible effect | Justified change |
|---|---|---|---|
| **H** | | | … because … |
| **G** | | | … because … |

## PART D Four-mark question




Record Q covers several months. Carrots supply beta-carotene, which the body converts to vitamin A. Explain why this is malnutrition, not starvation. State a possible effect and justify one change. **[4]**

**Revision:** code ___ ; the scientific relationship I changed is _____

**Help key:** N nutrient role · L deficiency link · B balance evidence · J justified change. Keep your usual access arrangements.